

WAGA FLARM : Firmware Update Instructions 21/05/26

1. Download the latest 'ATOM UAV Firmware' file from the FLARM website:



2. Make a note of where you have saved the file, eg in the root c:\. It is suggested, for convenience, to rename the file with a shorten name eg atom743.fw for atom firmware version 7.43.
3. You will need to have installed on your PC the Python 'pip' (Pip Installs Packages). This is an element of the Python coding packages. You can check whether you have this installed by going to the PC Command Line and typing 'pip help' Return. If it is installed, you will see an extensive text help file displayed. If is not installed, use AI to help you install it!
4. If not already downloaded and installed on your PC, download the 'WAGA FLARM firmware update wheel' file (flarm_serialutil-1.4.8-py3-none-any.whl) from the GitHub repository: github.com/Bumpff/SoftRF_Skyview-WAGA
5. From the PC Command Line, enter the following and type Return:

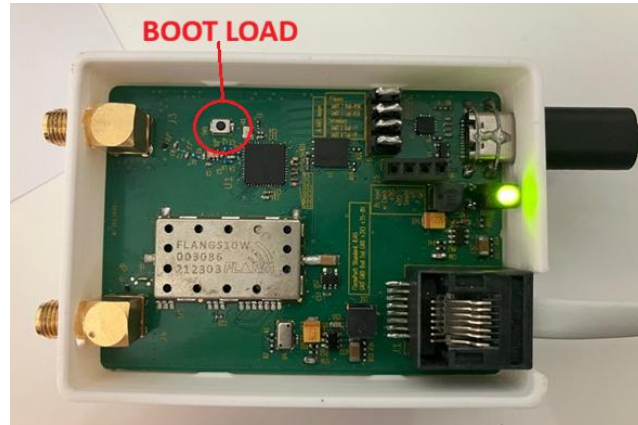
Pip install flarm_serialutil-1.4.8-py3-none-any.whl

This will effectively setup an installation programme.

6. Connect the WAGA Flarm USB port to the PC using a suitable data and power cable. Use PC Device Manager to see which port COM number you have connected to. If there is no connection, then 7 step will give an error.
7. From the PC Command Line (**change portCOM## to your COM number,**) initiate the update by entering this line followed by Return:

```
python -m serialutil --port COM11 update_fw_recovery c:\atom743.fw
```

8. Promptly after step 7 and whilst still in the startup process, you need to initiate the boot loader by either:
 - a. Press and release the boot load button (inside the WAGA FLARM) quickly, whilst the start-up sequence is in process in order to trigger the boot loader. Or;



- b. Unplug and replug in the connecting cable. This will recommence the start-up sequence.
9. The firmware will proceed to be installed in chunks as shown on the PC screen.
10. When complete the WAGA FLARM should reboot, or else force it to reboot by disconnecting the power.
11. Check the update has been successful using Putty or another PC terminal application, sending \$PFLAC,R, SWVER \r\n. The reply should be in the format: \$PFLAC,A,7.43.

Notes:

Understanding pip install 1 2 3



The `pip install` command is used to install Python packages from the Python Package Index (PyPI) or other package repositories. Pip stands for "Pip Installs Packages," and it is the standard package manager for Python. This command allows you to easily manage libraries and dependencies that are not included in the Python standard library.

Basic Usage

To install a package using `pip`, you can use the following syntax:

```
pip install <package_name>
```



For example, to install the `requests` library, you would run:

```
pip install requests
```



This command will download and install the latest version of the `requests` package from PyPI.

Python Wheel Format 1 2 3



A **Python Wheel** (`.whl`) is a **built distribution format** for Python packages, designed to make installations **faster, simpler, and more reliable** compared to source distributions (`.tar.gz`). It's essentially a **ZIP archive** with a specific naming convention that encodes compatibility details like Python version, ABI, and platform.

When you install a package with `pip`, if a compatible wheel is available, it skips the build step and installs directly — avoiding compilation and reducing dependency issues.

Key Benefits

- **Speed:** No compilation step; installs much faster.
- **Portability:** Prebuilt for specific Python versions and platforms.
- **No Compiler Needed:** Especially useful for packages with C/C++ extensions.
- **Consistency:** Avoids arbitrary code execution from `setup.py` during install.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\bumpf>pip help

Usage:
  pip <command> [options]

Commands:
  install          Install packages.
  download         Download packages.
  uninstall        Uninstall packages.
  freeze           Output installed packages in requirements format.
  inspect          Inspect the python environment.
  list             List installed packages.
  show             Show information about installed packages.
  check            Verify installed packages have compatible dependencies.
  config           Manage local and global configuration.
  search           Search PyPI for packages.
  cache            Inspect and manage pip's wheel cache.
  index            Inspect information available from package indexes.
  wheel            Build wheels from your requirements.
  hash             Compute hashes of package archives.
  completion       A helper command used for command completion.
  debug            Show information useful for debugging.
  help             Show help for commands.
```

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

```
C:\Users\bumpf>pip install c:\users\bumpf\flarm_serialutil-1.4.8-py3-none-any.whl
Processing c:\users\bumpf\flarm_serialutil-1.4.8-py3-none-any.whl
Collecting click (from flarm-serialutil==1.4.8)
  Downloading click-8.3.2-py3-none-any.whl.metadata (2.6 kB)
Collecting crcmod (from flarm-serialutil==1.4.8)
  Downloading crcmod-1.7.tar.gz (89 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting future (from flarm-serialutil==1.4.8)
  Downloading future-1.0.0-py3-none-any.whl.metadata (4.0 kB)
Collecting pyserial (from flarm-serialutil==1.4.8)
  Downloading pyserial-3.5-py2.py3-none-any.whl.metadata (1.6 kB)
Collecting six (from flarm-serialutil==1.4.8)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting colorama (from click->flarm-serialutil==1.4.8)
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Downloading click-8.3.2-py3-none-any.whl (108 kB)
Downloading future-1.0.0-py3-none-any.whl (491 kB)
Downloading pyserial-3.5-py2.py3-none-any.whl (90 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Building wheels for collected packages: crcmod
  Building wheel for crcmod (pyproject.toml) ... done
  Created wheel for crcmod: filename=crcmod-1.7-py3-none-any.whl size=18949 sha256=f4232794055ed58359e2f48bce4ffd7761cdbc24e99e66c1b4b7f66a5bdce925
  Stored in directory: c:\users\bumpf\appdata\local\pip\cache\wheels\c5\b4\7d\51d3485a8022501c7f7dba78b6c945088ff93d40142c9a96f4
Successfully built crcmod
Installing collected packages: pyserial, crcmod, six, future, colorama, click, flarm-serialutil
Successfully installed click-8.3.2 colorama-0.4.6 crcmod-1.7 flarm-serialutil-1.4.8 future-1.0.0 pyserial-3.5 six-1.17.0
```

```
C:\Users\bumpf>python -m serialutil --port COM4 update_fw_recovery c:\users\bumpf\atom743.fw
```

```
INFO:root:Waiting for bootloader
INFO:root:Bootloader ready
INFO:root:Found 1 entries in firmware
INFO:root:Transmitting: 0.00%
INFO:root:Transmitting: 0.07%
INFO:root:Transmitting: 0.14%
INFO:root:Transmitting: 0.21%
INFO:root:Transmitting: 0.28%
INFO:root:Transmitting: 0.36%
INFO:root:Transmitting: 0.43%
INFO:root:Transmitting: 0.50%
```

May need to press WAGA FLARM internal
bootloader press-button at this point